

***CADModelScope*: Revealing the Dependency Structure Behind Parametric Computer-Aided Design Models**

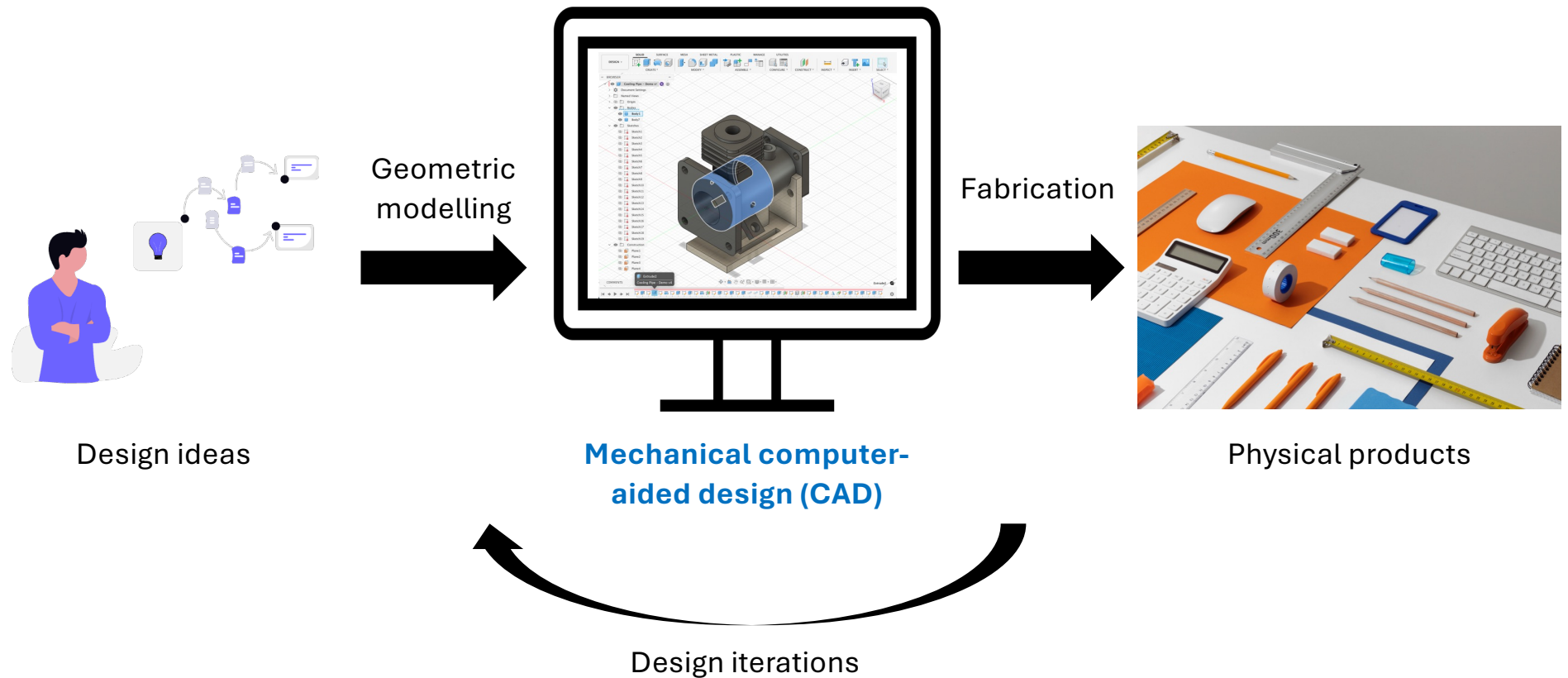
Yuanzhe Deng, Zhijing Zhang, Dr. Shurui Zhou & Dr. Alison Olechowski



UNIVERSITY OF
TORONTO



Modern product development



Dependencies in parametric CAD

Operation
Sequence



Sketch



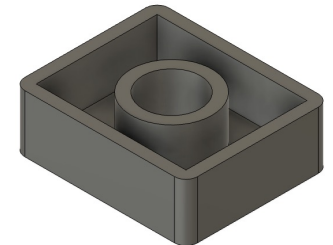
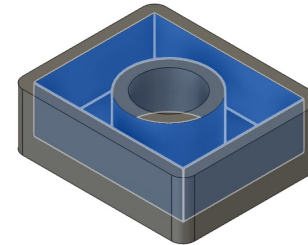
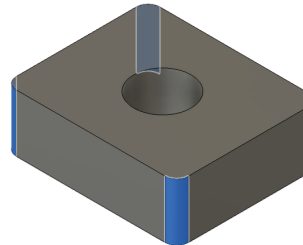
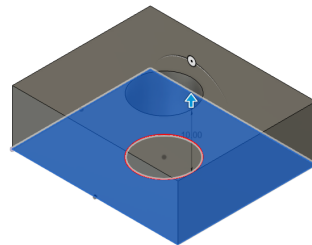
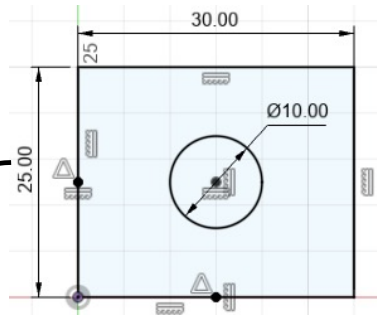
Extrude



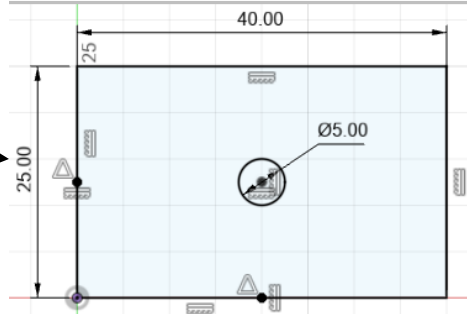
Fillet



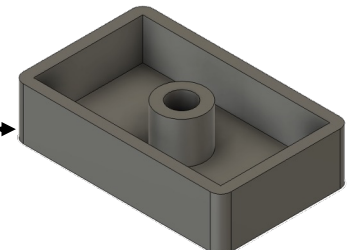
Shell



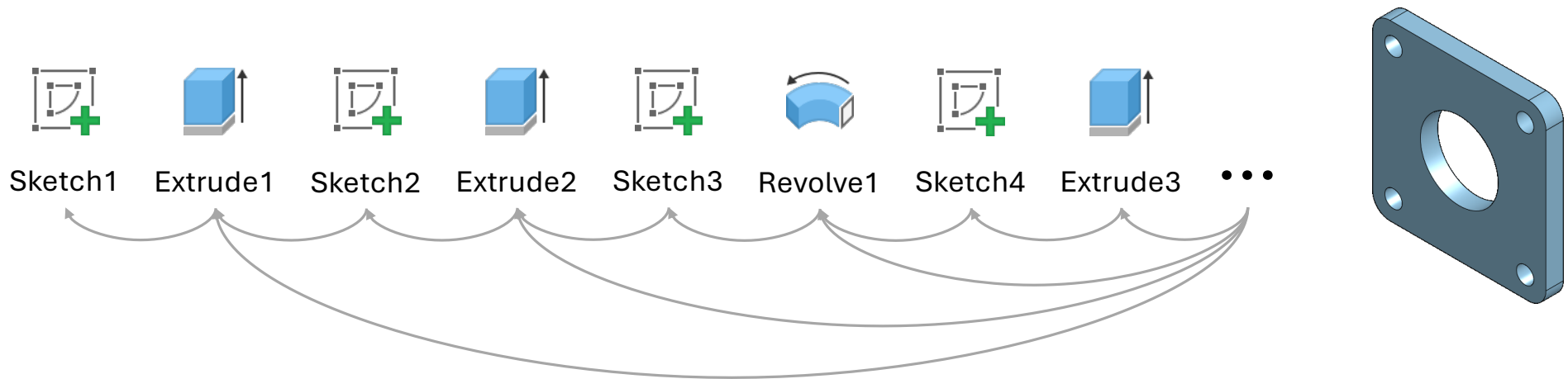
Edit

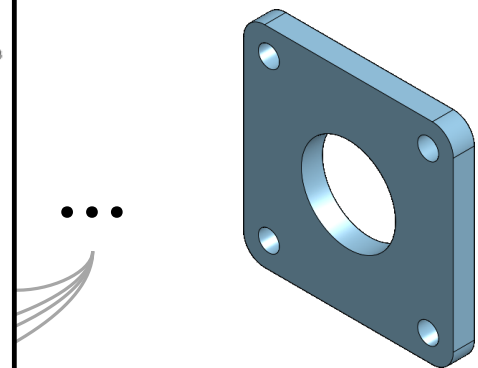
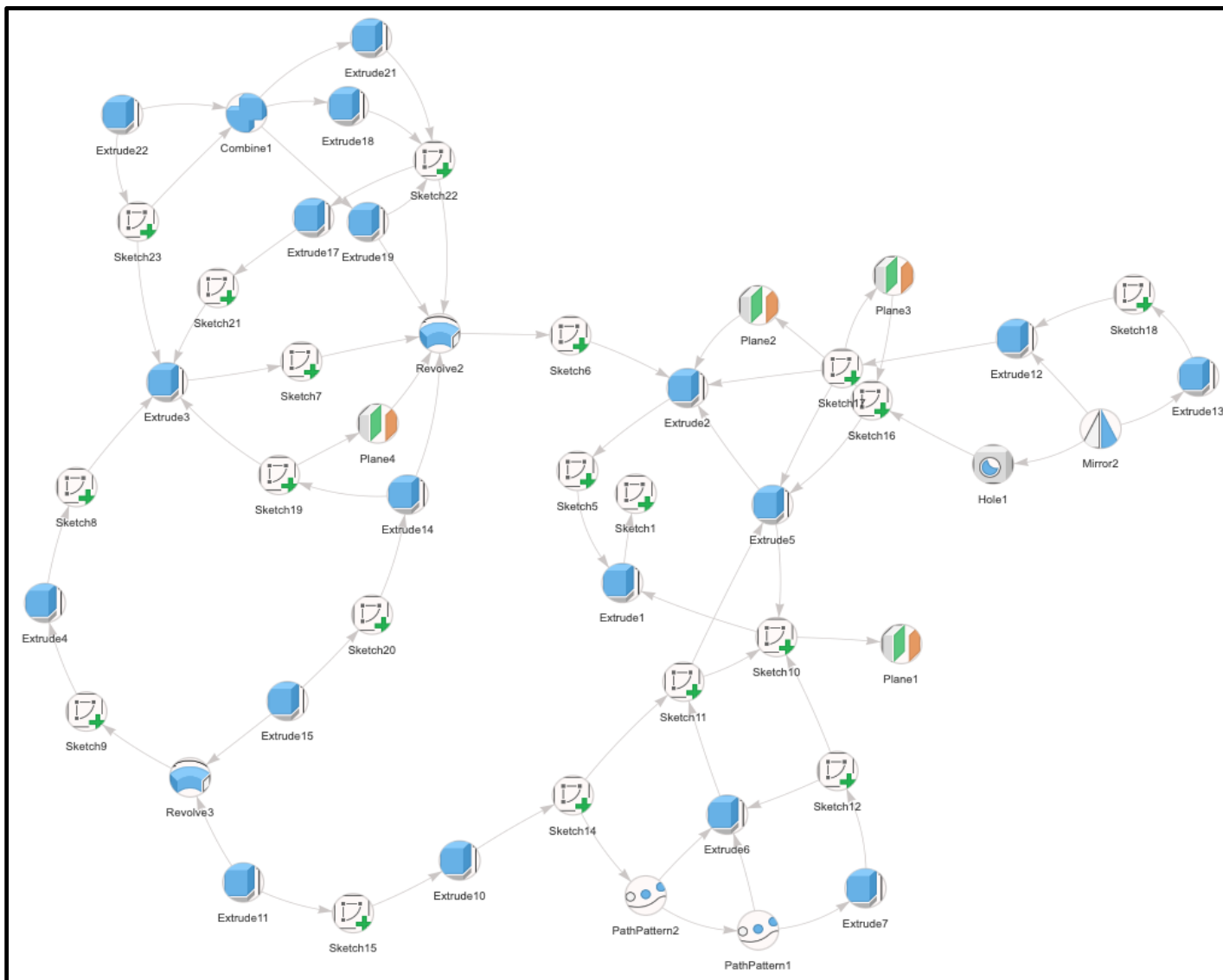


Automatic propagation



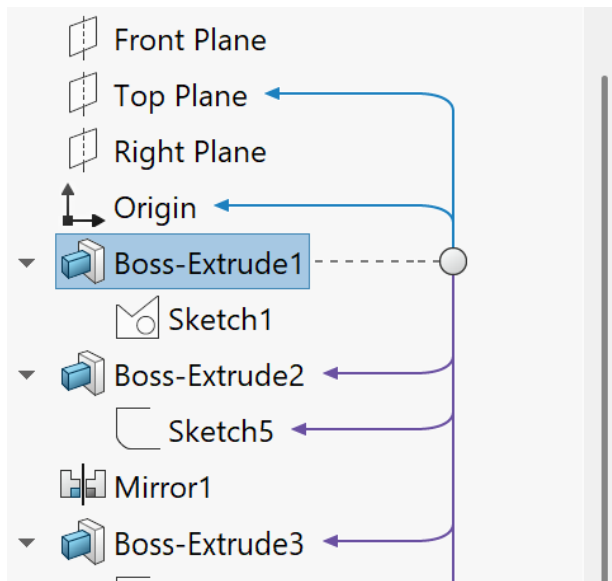
The underlying dependencies in CAD are complex and intricate!





**But this model is only
built with 48 operations!**

Dependency tracing in commercial CAD



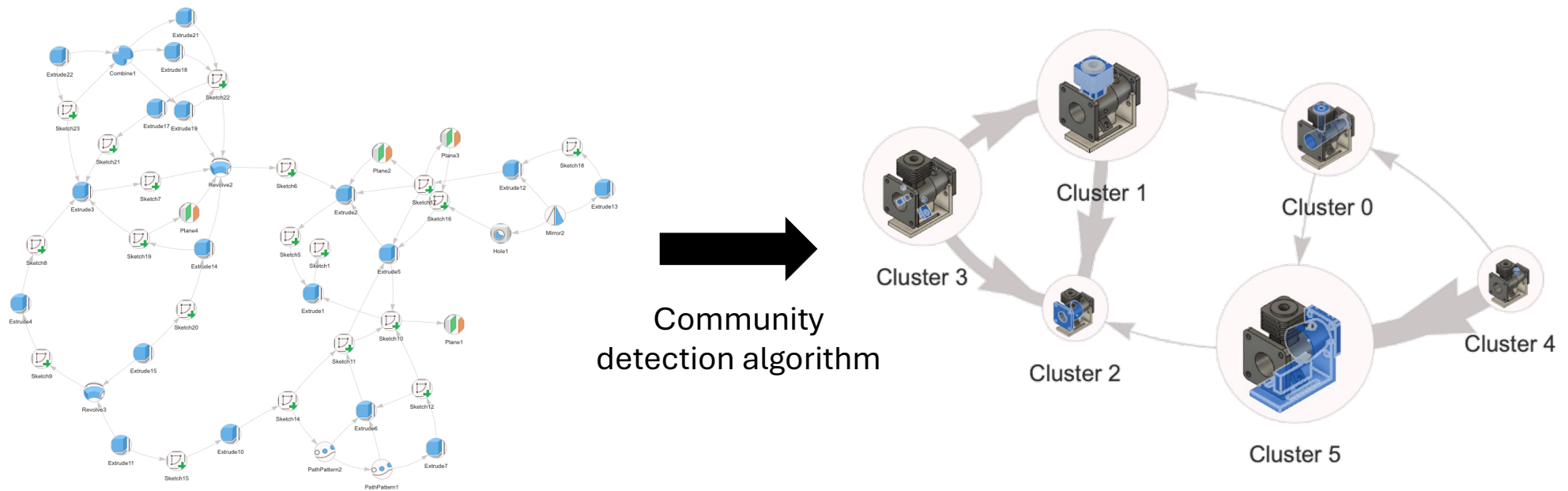
SolidWorks



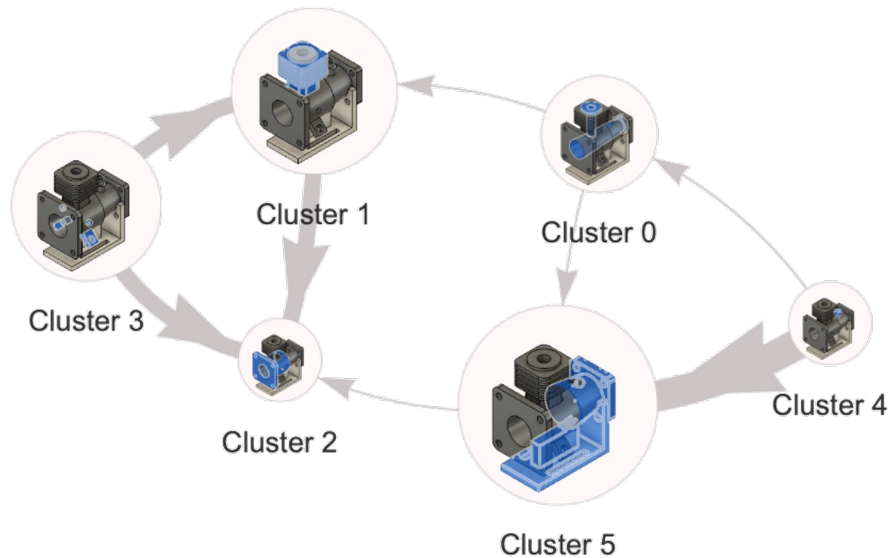
CATIA V5

Challenge: existing visualizations do not scale!

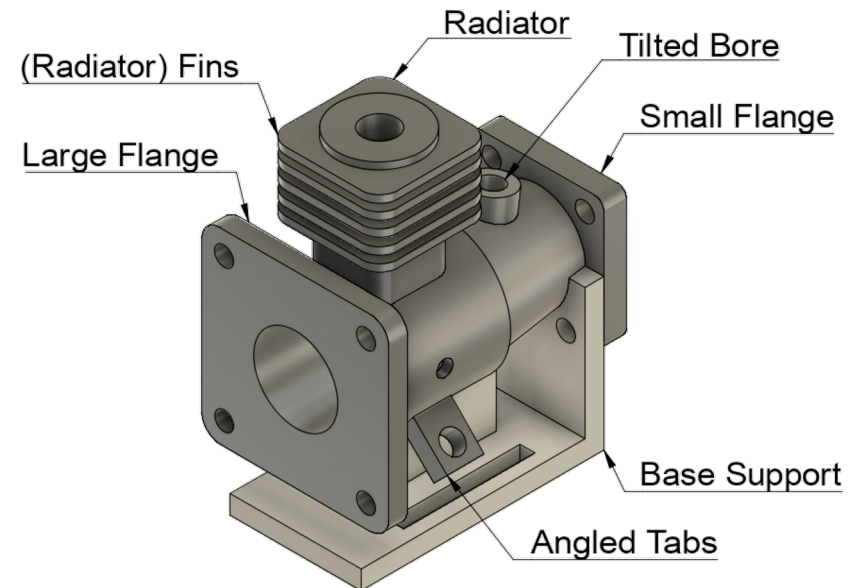
Operation modularization



Operation modularization

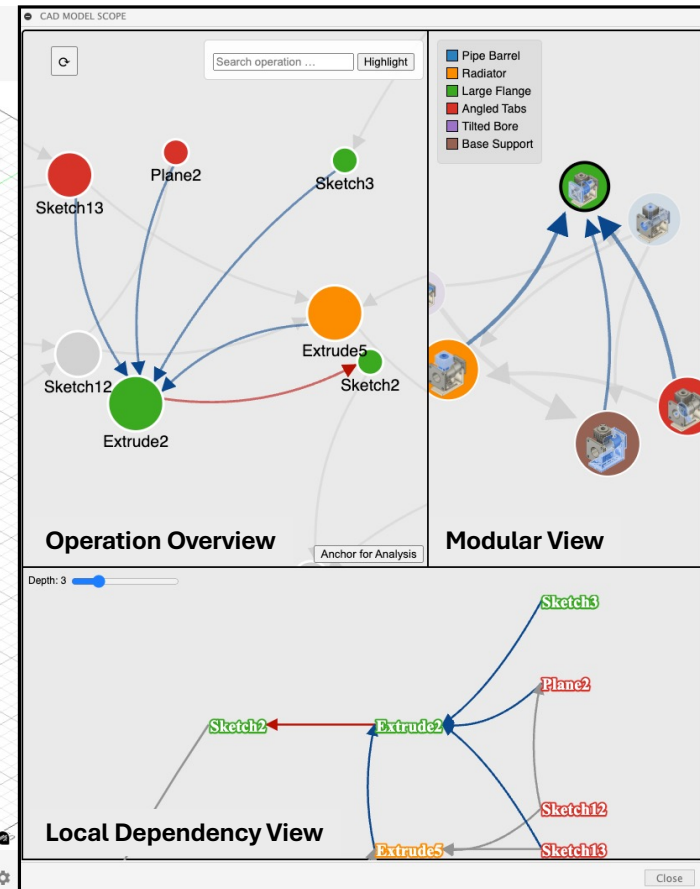
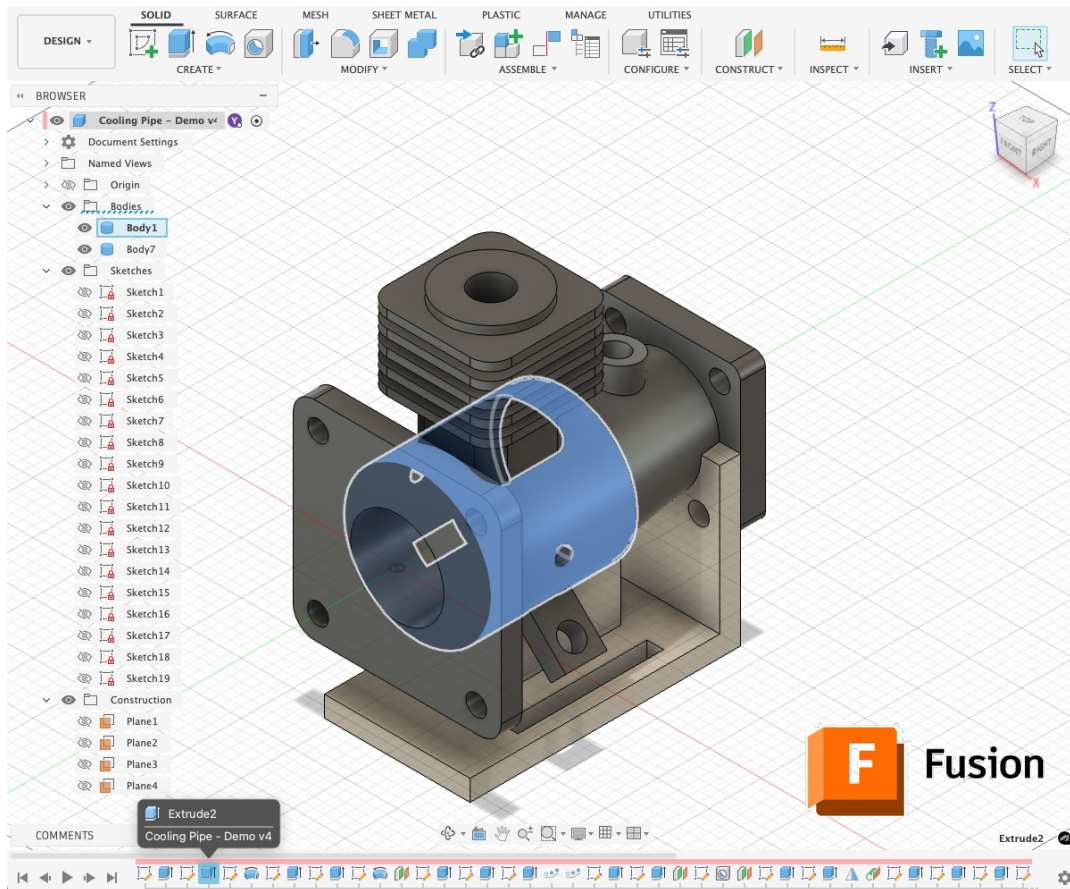


VS.



Y. Deng et al., What Sets Proficient and Expert Users Apart? Results of a Computer-Aided Design Experiment, *Journal of Mechanical Design*, 2024.

CADModelScope

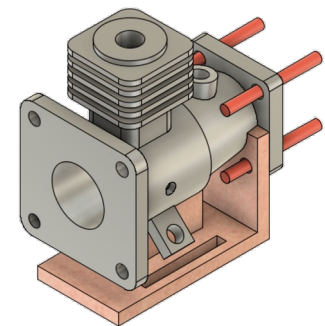


Evaluation study

ID [†]	Participation	CAD experience (years)	
		FUSION	Other CAD
A1	In-person	0	SOLIDWORKS (3)
A2	Virtual	5	INVENTOR (4), SOLIDWORKS (3)
A3	In-person	3	CATIA (2), SOLIDWORKS (1)
A4	Virtual	5	ONSHAPE (5), SOLIDWORKS (2), CREO (1)
B1	Virtual	3	INVENTOR (4), ONSHAPE (2), SOLIDWORKS (<1)
B2	In-person	1	SOLIDWORKS (3)
B3	In-person	0	SOLIDWORKS (2), ONSHAPE (2)
B4	In-person	3	None

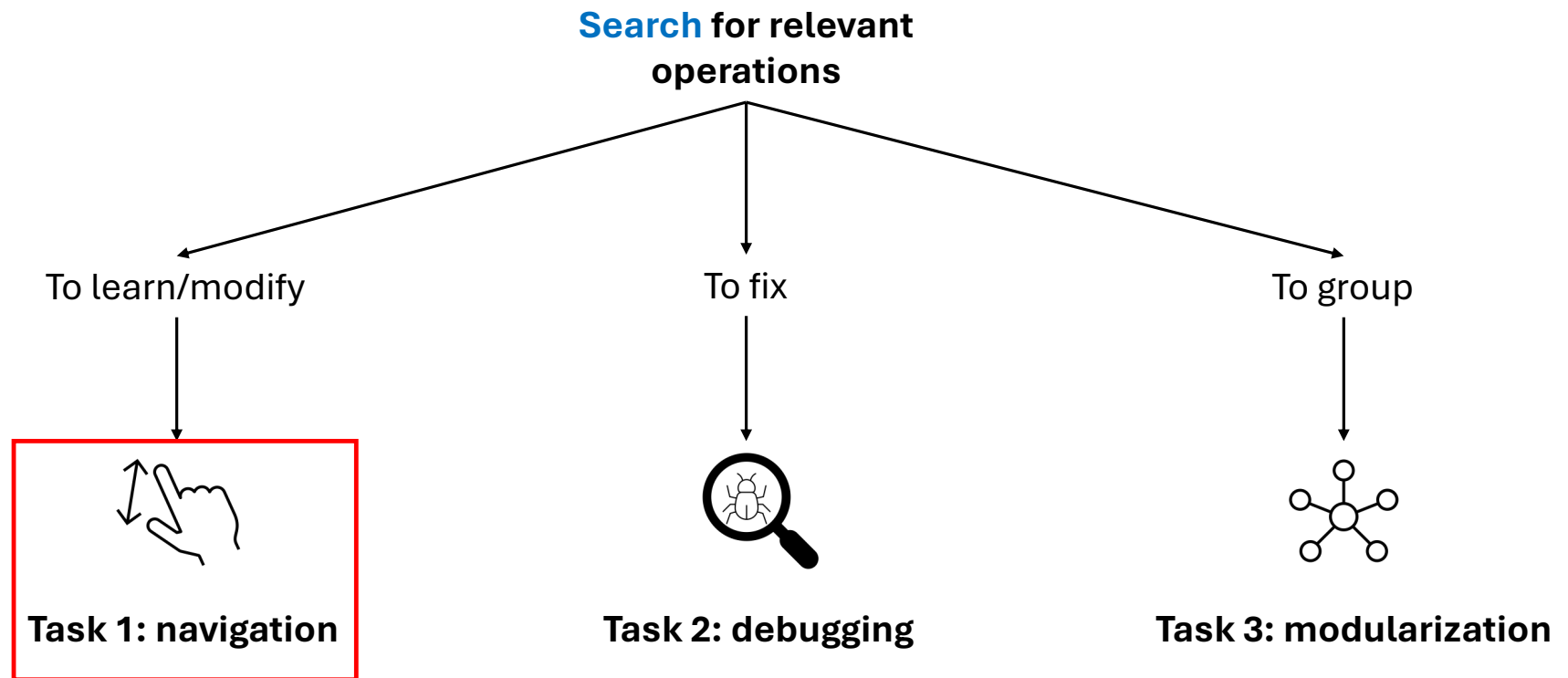


191 operations

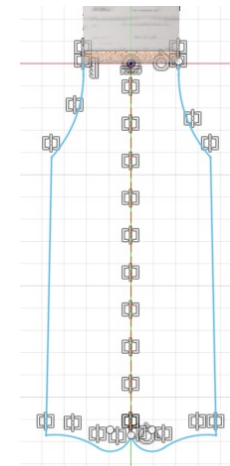
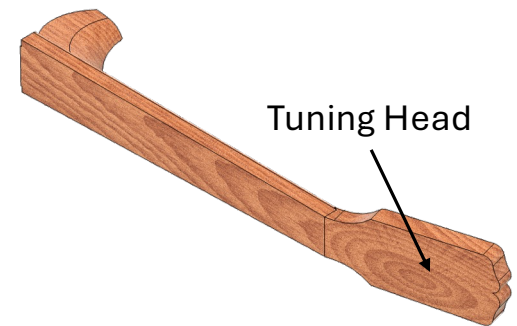
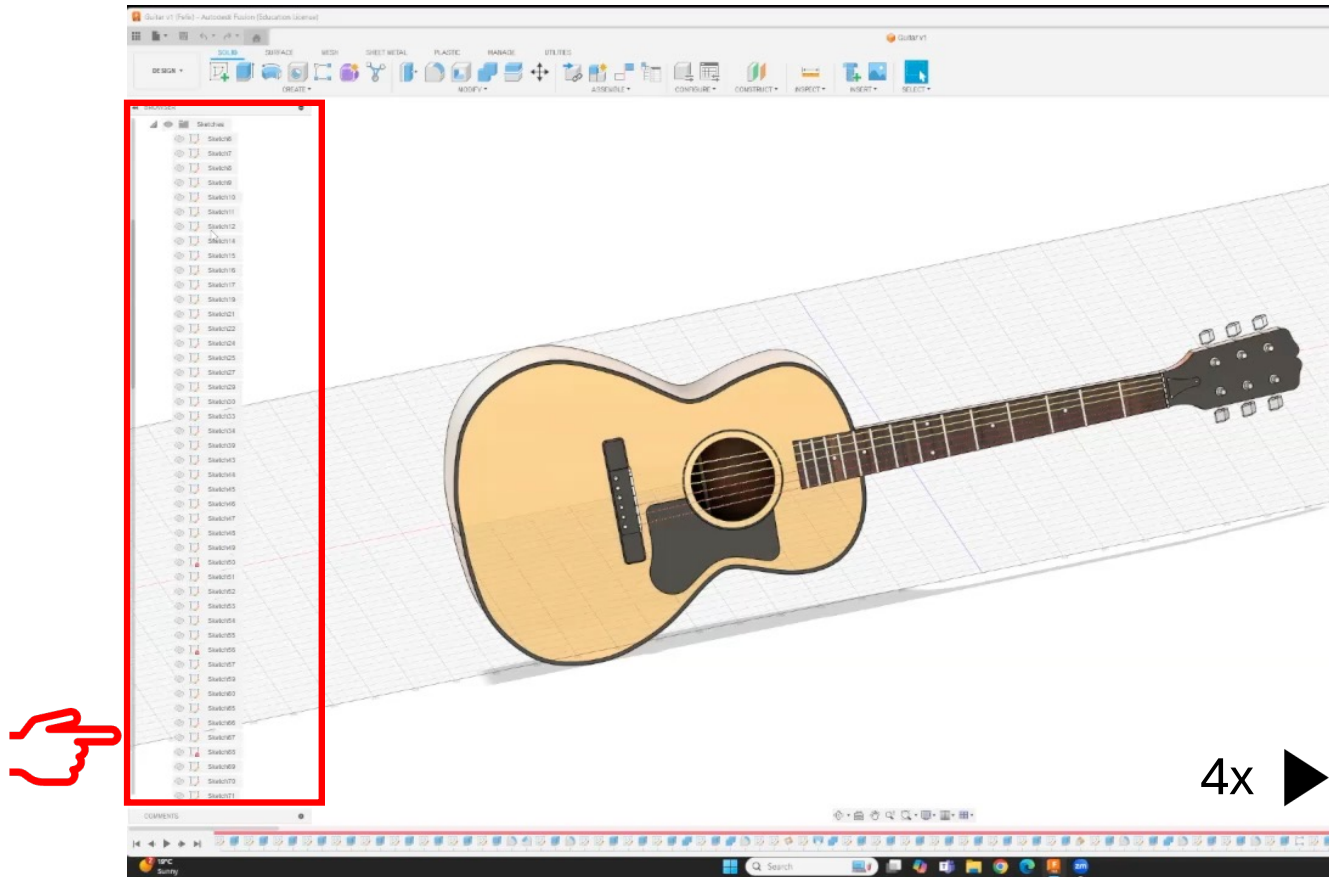


50 operations

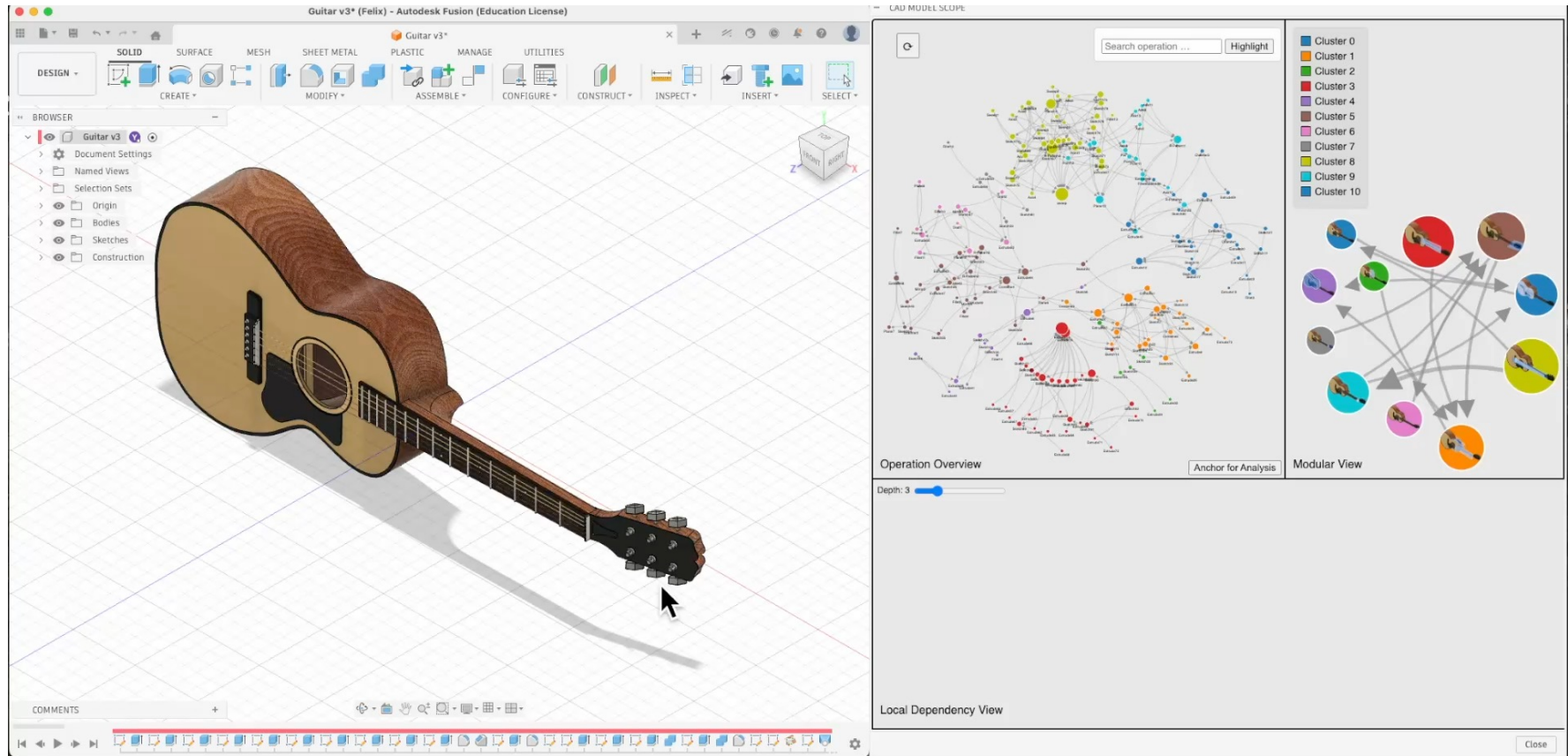
What novel strategies *does CADModelScope* enable?



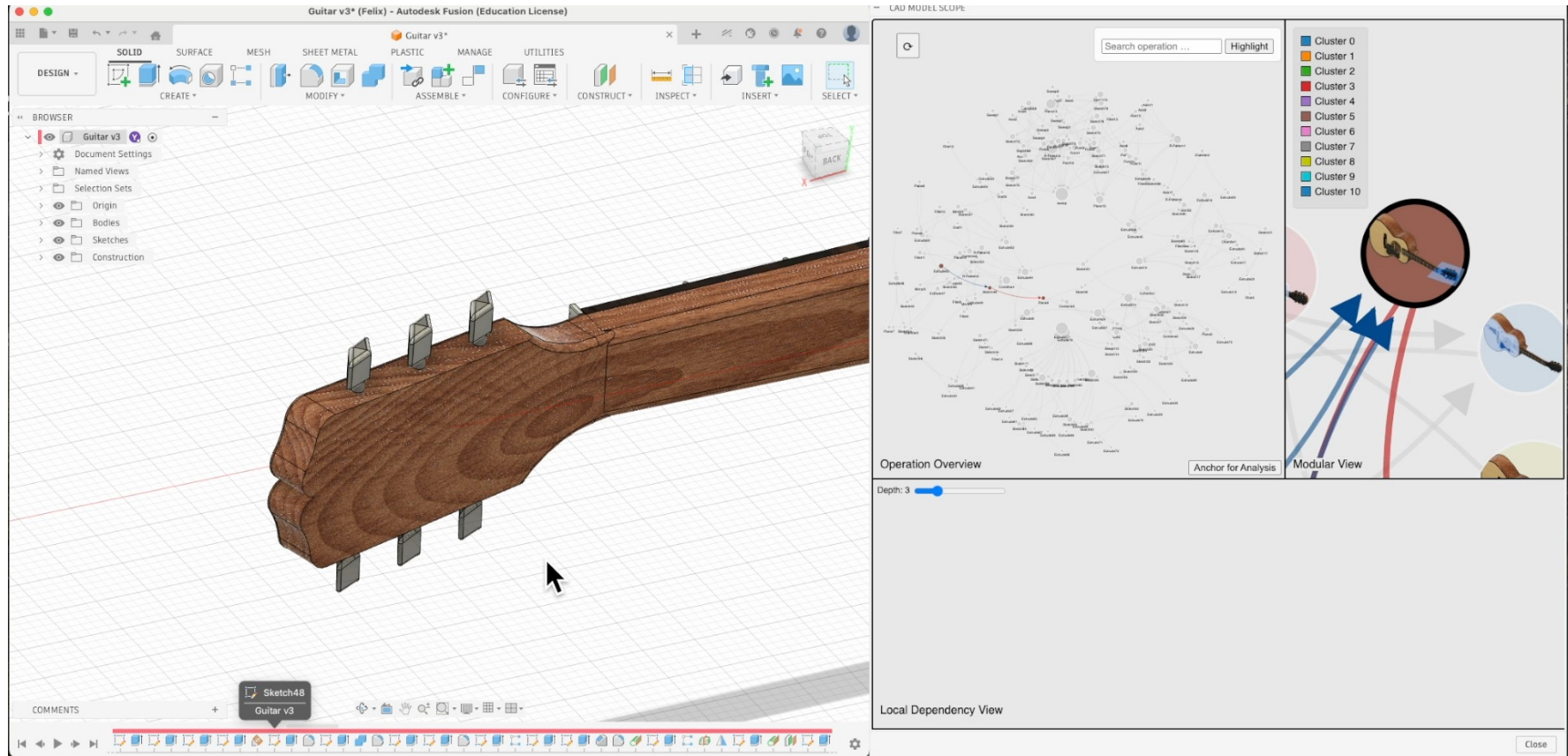
Without *CADModelScope*: exhaustive and error-prone search



CADModelScope: more structured navigation processes

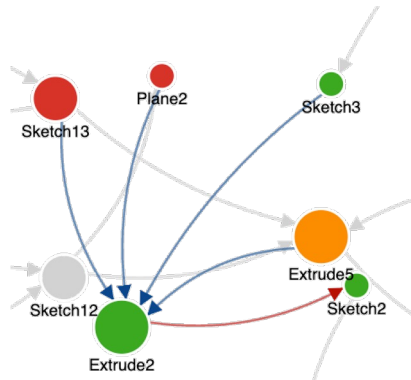


CADModelScope: more structured navigation processes



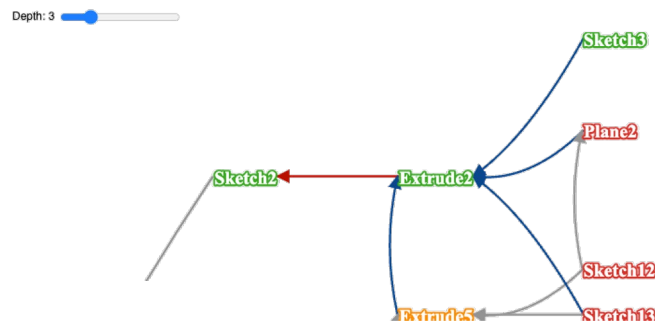
Design implications

Operation Overview



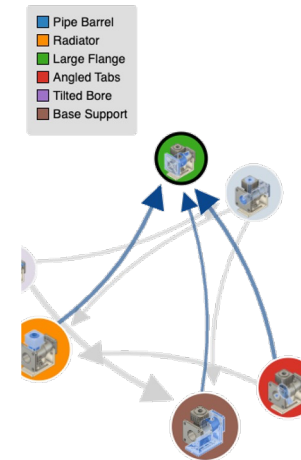
Raise user awareness of the underlying model representation

Local Dependency View



Juxtapose related operations by dependency

Modular View



Reduce graph complexity for information foraging

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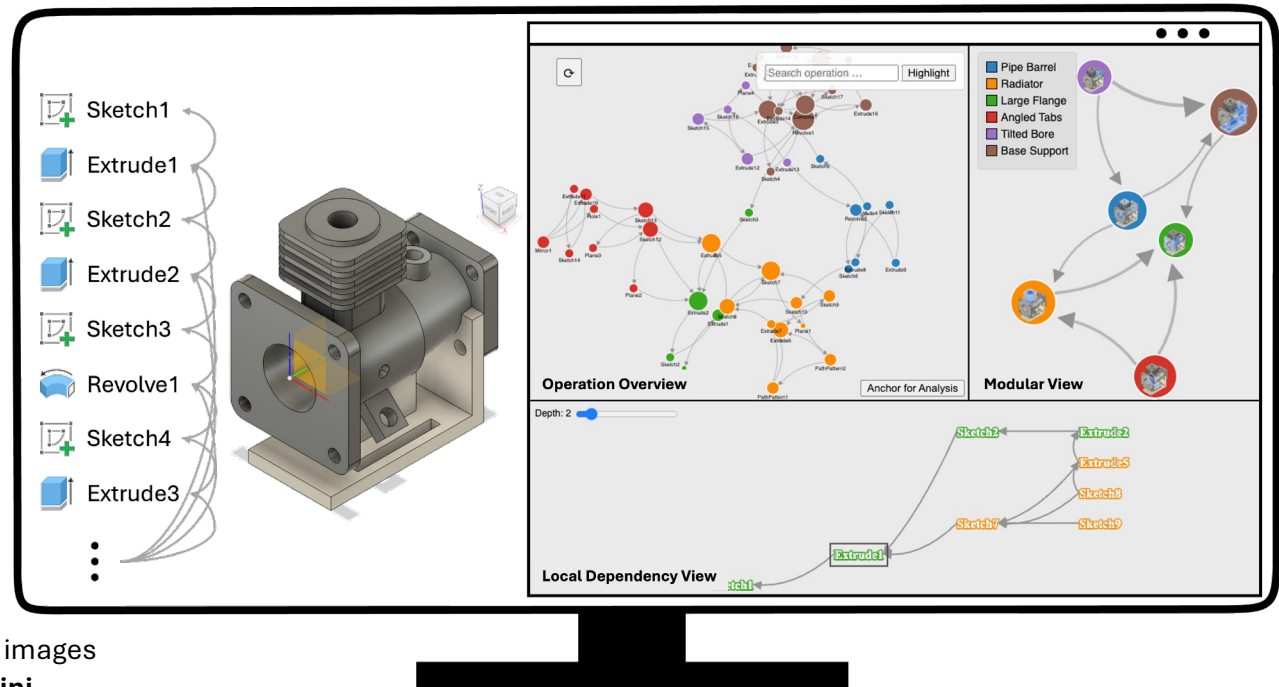
Contact: yuanzhe.deng@mail.utoronto.ca



Contact me!



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